Joint 3-D ISAR Imaging and Dynamic Estimation of Spinning Spacecraft Based on Signal Phase Analysis

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***Abstract*— Three-dimensional (3-D) geometry reconstruction and dynamic estimation of spinning spacecraft are crucial for space surveillance. Existing methods rely on interpreting 2-D inverse synthetic aperture radar (ISAR) images, which often suffer from image defocusing and scaling errors induced by target spin. To tackle this problem, we propose a joint 3-D ISAR imaging and dynamic estimation framework based on signal phase analysis, eliminating the need for prior 2-D imaging. After modeling the complex rotation of a spinning spacecraft relative to radar, we establish an overdetermined system of equations based on relationships between target parameters and cubic phase coefficients of radar echoes. Then, our two-step solution framework is organized as follows. First, we estimate cubic phase coefficients of the multicomponent signal through sparse representation (SR). To construct compact dictionaries for SR, we combine the coarse estimation and the phase order hierarchical processing (POHP) strategy to guide their design. Second, 3-D locations of scatterers, spin velocity, and direction of spin axis are estimated from the phase coefficients by solving the equation system. To ensure accurate and efficient solutions, we derive their iterative formulas using an alternating iterative strategy. With these formulas, 3-D ISAR imaging and dynamic estimation can be achieved directly. The effectiveness and superiority of our method are verified by experiments on simulated ISAR echoes.**

***Index Terms*— 3-D reconstruction, compressed sensing (CS), cubic phase signal, dynamic estimation, inverse synthetic aperture radar (ISAR) imaging, spinning space target.**

I. Introduction

TO ENSURE space security, monitoring spacecraft targets is essential [1], [2]. The analysis of their in-orbit statuses typically involves interpreting observation data from ground-based radars through the application of remote sensing technologies. When a spacecraft loses control, it struggles to maintain its triaxial stability and tends to enter a slow spin [3], [4]. In this situation, the estimation of its dynamic parameters, including spin velocity and direction of spin axis, is crucial. The dynamic estimation enables timely warnings about potential collisions caused by the spacecraft veering off its orbit [4], [5], [6], [7], [8], [9]. Additionally, 3-D geometry reconstruction can provide 3-D location parameters of target scatterers, which helps assess whether the target structure is deformed or damaged [3], [10], [11], [12], [13], [14], [15], [16], [17], [18].

Most existing methods for dynamic estimation and 3-D geometry reconstruction rely on the interpretation of 2-D inverse synthetic aperture radar (ISAR) images [3], [4], [5], [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18]. Dynamic estimation methods are typically divided into two categories. One category requires a preconstructed template library containing target images across all possible attitudes, and determines instantaneous attitudes and dynamic parameters of the target by matching measured images with template images [6]. Another category establishes geometric relationships among multiview images based on imaging projection theory. Dynamic parameters are then estimated by solving equations related to target projection features [7], [8], [9]. The 3-D geometry reconstruction methods are also classified into two types. In factorization-based methods, target scatterers are extracted and correlated across the image sequence to construct a scattering history matrix, and 3-D locations of scatterers are estimated by factorizing this matrix [10]. In imaging projection-based methods, the projection model for each image is determined by dynamic parameters, and the 3-D structure is rebuilt by judging whether a point is projected into the target areas of the multiview images [3], [11], [12], [13], [14], [15]. Notably, the performance of all these methods heavily relies on the quality of 2-D ISAR imaging. However, the complex rotation of a spinning spacecraft relative to radar is synthesized by its orbital motion and self-spin. The unknown spin complicates ISAR autofocusing and cross-range scaling [19], [20]. Consequently, it is difficult to extract the projection features or scatterers from defocused images, rendering the existing methods ineffective. Additionally, the scaling errors degrade the estimation accuracy.

To overcome the limitations imposed by the quality of 2-D ISAR imaging, directly estimating these target parameters from radar echoes presents a promising alternative. However, significant challenges persist in the estimation process. On the one hand, the connections between radar echoes and these target parameters remain unexplored. The total radar echo can be regarded as a multicomponent signal composed of the echoes from multiple strong scatterers, and the expressions

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for phase coefficients of each component need to be derived to establish the connections. On the other hand, the accuracy of phase coefficient estimation must be ensured. Sparse representation (SR) algorithms, also known as compressed sensing (CS) algorithms, are widely applied in multicomponent signal decomposition due to their robustness against noise [18], [21], [22], [23], [24]. However, SR requires an overcomplete dictionary, whose size increases dramatically when the planar scatterers are extended to 3-D scatterers. The large-scale dictionary entails huge storage cost and low computational efficiency [25], necessitating simplification without compromising its completeness.

In this article, we propose a framework of joint 3-D ISAR imaging and dynamic estimation for spinning spacecraft based on signal phase analysis. By modeling the complex rotation of a spinning spacecraft in a coordinate system defined by the spin axis direction, we establish an overdetermined system of equations that describes the connections between cubic phase coefficients and target parameters. Then, our two-step method is outlined as follows: First, radar echoes are regarded as multicomponent cubic phase signals, and we estimate the phase coefficients by SR. To enhance the efficiency of SR, we employ a phase order hierarchical processing (POHP) strategy to design more compact dictionaries based on coarse estimates of phase coefficients obtained from a generalized cubic phase function (GCPF) algorithm [26]. Second, target parameters are estimated from phase coefficients by solving the complex system of equations. To avoid the high-dimensional searches, an alternating iterative strategy is employed along with the introduction of intermediate parameters to derive iterative formulas for the solution. To validate the effectivity and superiority of our method, we conduct simulation experiments on spacecraft under different spinning states. The main contributions are summarized below.

1) We propose a joint 3-D ISAR imaging and dynamic estimation framework for spinning spacecraft based on signal phase analysis. Our framework estimates location parameters of scatterers and dynamic parameters directly from echoes, rather than from 2-D images, bypassing the issues of image defocusing and cross-range scaling errors.
2) In phase coefficient estimation based on SR, we integrate coarse estimation with a POHP strategy to design more compact dictionaries. This compact dictionary allows for accurate SR with low storage requirements and computational costs.
3) To estimate target parameters from phase coefficients, we employ an alternating iterative strategy and introduce intermediate parameters, enabling accurate and rapid solutions to the complex overdetermined system of equations.

The remainder of this article is organized as follows. In Section II, the 3-D ISAR imaging model of spinning spacecraft is established. In Section III, the proposed framework is introduced along with details about the dictionary design and the alternating iterative algorithm. In Section IV, experimental results are presented. In Section V, a brief conclusion is addressed.

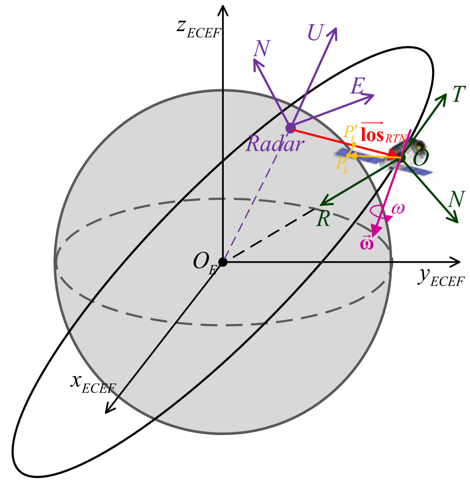

Fig. 1. ISAR observation geometry of a spinning spacecraft.

II. 3-D ISAR Imaging Model of Spinning Spacecraft

As illustrated in Fig. 1, the rotation of a slowly spinning spacecraft relative to the ground-based radar consists of two parts, which are, respectively, caused by its orbital motion and spin. The rotation caused by orbital motion is equivalent to the rotation of the radar line of sight (LOS) in a local orbital coordinate system, named the RTN system [27]. The axes of the RTN system are defined by radial (R), along-track (T), and cross-track (N) directions. The angular acceleration of its spin is determined by the moment of inertia and disturbance torque. According to the relevant research on satellite attitude control [28], [29], the angular acceleration is typically less than 10^{-5} rad/s^2. Therefore, the spin axis and velocity of the spacecraft can be considered fixed during a short imaging coherent processing interval (CPI) (within 10 s) [3], [7]. During the CPI, M pulses are transmitted and received by radar. The slow time $t = m/\text{prf}$, $m = 0, 1, \ldots, M-1$, prf is the pulse repetition frequency (PRF). For the kth target scatterer (labeled as P_k in figures), its initial location in the RTN system is $\vec{\mathbf{p}}_{\text{RTN},k} \in \mathcal{R}_{3\times 1}$, and its distance $R_k(t)$ to the radar changes with slow time

$$\begin{aligned} R_k(t) &\approx R_{RO}(t) + R_{OP',k}(t) \\ &= R_{RO}(t) + \overrightarrow{\mathbf{los}}_{\text{RTN},t}^{\text{T}} \left(\mathbf{Rot}(\alpha, \beta, \omega \cdot t)\, \vec{\mathbf{p}}_{\text{RTN},k}\right) \end{aligned} \quad (1)$$

where $R_{RO}(t)$ is the distance between radar and target centroid; $\overrightarrow{\mathbf{los}}_{\text{RTN},t} \in \mathcal{R}_{3\times 1}$ is the instantaneous LOS vector in the RTN system, which is derived from radar tracking information through coordinate transformation [30]; P'_k is the projection of P_k on the LOS, $R_{OP',k}(t)$ is the distance between O and P'_k; $\mathbf{Rot}(\alpha, \beta, \omega \cdot t)$ is the spin matrix determined by dynamic parameters, α and β are azimuth and elevation angles of the spin axis in the RTN system, the direction of spin axis is expressed as $\vec{\boldsymbol{\omega}} = [\cos\beta\cos\alpha, \cos\beta\sin\alpha, \sin\beta]^{\text{T}}$, and ω is the spin velocity.

The radar echo after pulse compression, translation compensation (TC) [31], and migration through resolution cell

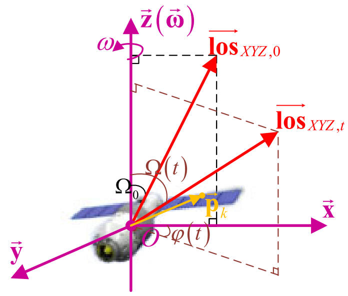

Fig. 2. Relative rotation of a spinning spacecraft in the *XYZ* coordinate system we defined.

correction (MTRCC) [32] can be represented as follows:

$$E(\tau,t)=\sum_{k=1}^{K}E_k(\tau,t)$$
$$E_k(\tau,t)=\sigma_k\cdot\sin c\left(B\left(\tau-\frac{2R_{OP',k}(0)}{c}\right)\right)\cdot\exp\left(-j\frac{4\pi}{c}f_cR_{OP',k}(t)\right) \quad (2)$$

where τ is the fast time, σ_k is the scatterer amplitude, c is the speed of light, B is the bandwidth, and f_c is the carrier frequency.

Considering the form complexity of spin matrix, we define a target coordinate system (XYZ system) to simplify the expressions for $R_{OP',k}(t)$. As shown in Fig. 2, its origin is set at target centroid O, and the axis directions are defined as follows:

$$\begin{cases}\vec{\mathbf{x}}=\overrightarrow{\mathbf{los}}_{\text{RTN},0}-\vec{\boldsymbol{\omega}}\overrightarrow{\mathbf{los}}_{\text{RTN},0}^{\text{T}}\vec{\boldsymbol{\omega}}, & \vec{\mathbf{x}}=\vec{\mathbf{x}}/\left\|\vec{\mathbf{x}}\right\|_2\\ \vec{\mathbf{z}}=\vec{\boldsymbol{\omega}} & \\ \vec{\mathbf{y}}=\vec{\mathbf{x}}\otimes\vec{\mathbf{z}}, & \vec{\mathbf{y}}=\vec{\mathbf{y}}/\left\|\vec{\mathbf{y}}\right\|_2\end{cases} \quad (3)$$

where $\overrightarrow{\mathbf{los}}_{\text{RTN},0}$ is the LOS vector at time $t=0$, $\otimes$ represents the cross product of two vectors, $\|\cdot\|_2$ represents l-2 norm of the vector.

The LOS vector in the XYZ system is represented as follows:

$$\overrightarrow{\mathbf{los}}_{XYZ,t}=\left[\cos\varphi(t)\sin\Omega(t),\sin\varphi(t)\sin\Omega(t),\cos\Omega(t)\right]^{\text{T}} \quad (4)$$

where $\Omega(t)$ is the angle between $\overrightarrow{\mathbf{los}}_{\text{RTN},t}$ and $\vec{\boldsymbol{\omega}}$, $\varphi(t)$ is the angle between $\overrightarrow{\mathbf{lp}}_t$ and $\vec{\mathbf{x}}$, $\overrightarrow{\mathbf{lp}}_t$ is the projection of $\overrightarrow{\mathbf{los}}_{\text{RTN},t}$ on the "XY" plane, given by the following equation:

$$\overrightarrow{\mathbf{lp}}_t=\frac{\overrightarrow{\mathbf{los}}_{\text{RTN},t}-\vec{\boldsymbol{\omega}}\overrightarrow{\mathbf{los}}_{\text{RTN},t}^{\text{T}}\vec{\boldsymbol{\omega}}}{\left\|\overrightarrow{\mathbf{los}}_{\text{RTN},t}-\vec{\boldsymbol{\omega}}\overrightarrow{\mathbf{los}}_{\text{RTN},t}^{\text{T}}\vec{\boldsymbol{\omega}}\right\|_2}. \quad (5)$$

We define $\vec{\mathbf{p}}_k=[x_k,y_k,z_k]^{\text{T}}$ as the initial location of the kth scatterer in the XYZ system. During the spin, its locations can be expressed as follows:

$$\vec{\mathbf{p}}_{k,t}=\begin{bmatrix}\cos\omega t & -\sin\omega t & 0\\ \sin\omega t & \cos\omega t & 0\\ 0 & 0 & 1\end{bmatrix}\begin{bmatrix}x_k\\ y_k\\ z_k\end{bmatrix}. \quad (6)$$

Then, the distance $R_{OP',k}(t)$ can be simplified as follows:

$$R_{OP',k}(t)=\overrightarrow{\mathbf{los}}_{XYZ,t}^{\text{T}}\vec{\mathbf{p}}_{k,t}=R_{z,k}(t)+R_{xy,k}(t) \quad (7)$$
$$R_{z,k}(t)=\cos\Omega(t)\cdot z_k \quad (8)$$
$$R_{xy,k}(t)=\sin\Omega(t)\cdot\left(\cos\left(\varphi(t)-\omega t\right)\cdot x_k+\sin\left(\varphi(t)-\omega t\right)\cdot y_k\right). \quad (9)$$

We make the following approximations:

$$\Omega(t)\approx\Omega_o+\varepsilon_1\cdot t,\ \varphi(t)\approx\varepsilon_2\cdot t \quad (10)$$
$$\cos\omega t\approx 1-\omega^2t^2/2,\quad \sin\omega t\approx\omega t-\omega^3t^3/6. \quad (11)$$

We ensure the accuracy of approximation (10) by selecting the CPI, where observation angles vary linearly. The approximation (11) requires that the rotational angle $\omega\cdot t$ should be smaller than $\pi/3$. Then, we arrange $R_{OP',k}(t)$ into a cubic polynomial with respect to slow time

$$R_{OP',k}(t)\approx\left(\vec{\mathbf{h}}_0+\vec{\mathbf{h}}_1\cdot t+\vec{\mathbf{h}}_2\cdot t^2+\vec{\mathbf{h}}_3\cdot t^3\right)\vec{\mathbf{p}}_k \quad (12)$$
$$\vec{\mathbf{h}}_0=\left[\sin\Omega_o\quad 0\quad \cos\Omega_o\right] \quad (13)$$
$$\vec{\mathbf{h}}_1=\left[\varepsilon_1\cos\Omega_o\quad -\omega_e\sin\Omega_o\quad -\varepsilon_1\sin\Omega_o\right] \quad (14)$$
$$\vec{\mathbf{h}}_2=-\frac{1}{2}\left[\left(\varepsilon_1^2+\omega_e^2\right)\sin\Omega_o\quad 2\omega_e\varepsilon_1\cos\Omega_o\quad \varepsilon_1^2\cos\Omega_o\right] \quad (15)$$
$$\vec{\mathbf{h}}_3=\frac{1}{6}\begin{bmatrix}-\left(3\omega_e^2\varepsilon_1+\varepsilon_1^3\right)\cos\Omega_o\\ \left(3\omega_e\varepsilon_1^2+\omega_e^3\right)\sin\Omega_o\\ \varepsilon_1^3\sin\Omega_o\end{bmatrix}^{\text{T}} \quad (16)$$

where $\omega_e=\omega-\varepsilon_2$ is the effective spin velocity.

For a target with K strong scatterers, three dynamic parameters and $3K$ location parameters of scatterers need to be estimated. To achieve this, we establish the following overdetermined system of equations to estimate $3K+3$ target parameters:

$$\Xi=-\frac{2}{c}f_c\cdot\mathbf{H}(\alpha,\beta,\omega)\left[\vec{\mathbf{p}}_1\quad\vec{\mathbf{p}}_2\quad\cdots\quad\vec{\mathbf{p}}_k\right]_{3\times K}$$
$$\Xi=\begin{bmatrix}\xi_{0,1} & \xi_{0,2} & \cdots & \xi_{0,K}\\ \xi_{1,1} & \xi_{1,2} & \cdots & \xi_{1,K}\\ \xi_{2,1} & \xi_{2,2} & \cdots & \xi_{2,K}\\ \xi_{3,1} & \xi_{3,2} & \cdots & \xi_{3,K}\end{bmatrix}_{4\times K}$$
$$\mathbf{H}(\alpha,\beta,\omega)=\begin{bmatrix}\vec{\mathbf{h}}_0\\ \vec{\mathbf{h}}_1\\ \vec{\mathbf{h}}_2\\ \vec{\mathbf{h}}_3\end{bmatrix}_{4\times 3} \quad (17)$$

where Ξ is the phase coefficient matrix. For the kth scatterer, $\xi_{0,k}$, $\xi_{1,k}$, $\xi_{2,k}$, and $\xi_{3,k}$ are respectively the zero-order, linear, quadratic, and cubic phase coefficients of its azimuth echoes. $\mathbf{H}(\alpha,\beta,\omega)$ is the dynamic matrix. Then, the radar echo of the kth scatterer is represented as follows:

$$E(\tau,t)=\sigma_k\cdot\sin c\left(B\left(\tau+\frac{\xi_{0,k}}{f_c}\right)\right)\cdot\exp\left(j2\pi\left(\xi_{0,k}+\xi_{1,k}t+\xi_{2,k}t^2+\xi_{3,k}t^3\right)\right). \quad (18)$$

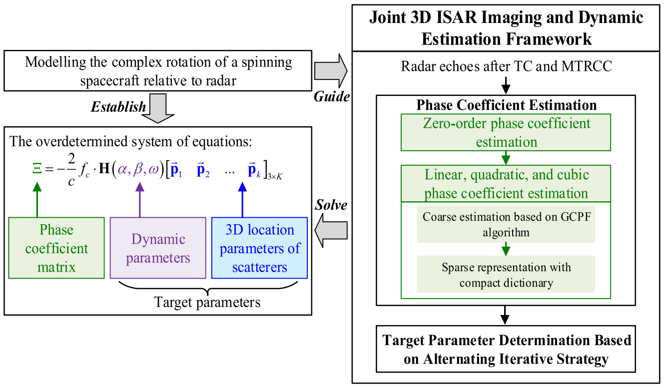

Fig. 3. Diagram of our joint 3-D ISAR imaging and dynamic estimation framework.

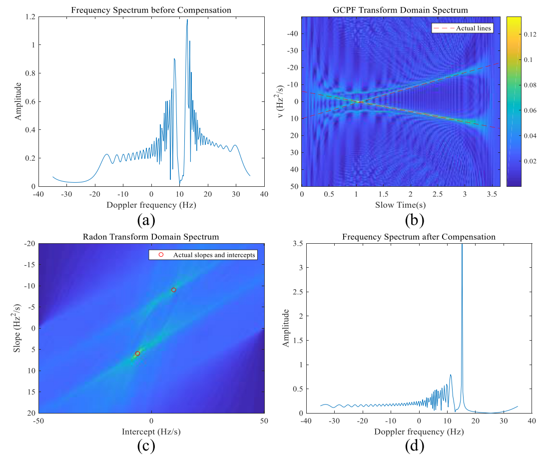

Fig. 4. Spectra of a two-component signal whose phase coefficients are $\xi_{1,1} = 15$, $\xi_{1,2} = -3$, $\xi_{1,3} = 1$ and $\xi_{2,1} = 3$, $\xi_{2,2} = 5$, $\xi_{2,3} = -1.5$. (a) Frequency spectrum before compensation. (b) GCPF transform domain spectrum. (c) Radon transform domain spectrum. (d) Frequency spectrum after compensating for the high-order phases of the 1st component. In (b) and (c) the actual lines and their parameters are marked in red.

III. 3-D ISAR Imaging and Dynamic Estimation Based on Signal Phase Analysis

A. Method Framework

As shown in Fig. 3, our framework accomplishes 3-D ISAR imaging and dynamic estimation by conducting the phase coefficient estimation and target parameter determination in succession. In phase coefficient estimation, we use SR with a compact dictionary. In target parameter determination, we design an alternating iterative algorithm.

B. Phase Coefficient Estimation Based on SR

1) Zero-Order Phase Coefficient Estimation: After TC and MTRCC, the azimuth echoes of a single scatterer are aligned in the range direction. Then, the range cells, which contain strong scatterers, can be selected by the 1-D orthogonal matching pursuit (OMP) algorithm [33], eliminating the effect of noise and sidelobes.

For a scatterer whose signal peak is located in the nth range cell, its zero-order phase coefficient $\xi_{0,k}$ can be estimated using by the following equation:

$$\hat{\xi}_{0,k} \approx -\frac{2f_c}{c} \cdot \left(\frac{c}{2B}n\right) = -\frac{f_c n}{B}. \tag{19}$$

After compensating for these phases, its signal can be expressed as follows:

$$E_k(t) \approx \sigma_k \cdot \exp\left(j2\pi\left(\xi_{1,k}t + \xi_{2,k}t^2 + \xi_{3,k}t^3\right)\right). \tag{20}$$

2) Linear, Quadratic, and Cubic Phase Coefficient Estimation: For each selected range cell, we conduct the multicomponent signal decomposition to estimate the linear, quadratic, and cubic phase coefficients. First, a coarse estimation is achieved using the GCPF algorithm [26]. Then, a more precise estimation is achieved by SR. To improve its efficiency, we employ a POHP strategy to design the compact dictionary for SR with reference to coarse estimates.

a) Coarse estimation based on the GCPF algorithm: The popular algorithms for estimating high-order phase coefficients include the Wigner–Ville distribution (WVD) [34], the matched Fourier transform [35], and the cubic phase function (CPF) [20], [26], [36]. Among these algorithms, the CPF can quickly process multicomponent signals in combination with the CLEAN technique [37]. We employ a GCPF algorithm [26], which modifies the CPF kernel function as follows:

$$\begin{aligned}\mathcal{G}_{E_k}(t,\nu) &= \int_0^\infty E_k(t+t_d)\,E_k(t-t_d)\,E_k^*(t)\,E_k^*(t) \\ &\quad \times e^{-j2\pi\nu t_d^2}\,dt_d \\ &= \sigma_k^4 \int_0^\infty e^{j2\pi t_d^2(2\xi_{2,k}+6\xi_{3,k}t-\nu)}\,dt_d \end{aligned} \tag{21}$$

where $E_k^*(t)$ is the conjugate of $E_k(t)$.

The frequency spectra and transform domain spectra of a two-component signal are illustrated in Fig. 4. In the spectrum of the GCPF transform domain, the energy of the signal is concentrated along the line: $\nu = 2\xi_{2,k} + 6\xi_{3,k}t$. The phase coefficients $\hat{\xi}_{c,2,k}$ and $\hat{\xi}_{c,3,k}$ can be determined from the line slope and intercept of this line, which are estimated using the Radon transform. After phase compensation and the Fourier transform, the frequency-domain signal is expressed as follows:

$$E'_k(f) \approx \sigma_k \delta\left(f - \xi_{1,k}\right) \tag{22}$$

where $\delta(\cdot)$ is the impulse function, and $\hat{\xi}_{c,1,k}$ is determined by searching for the peak of $E'_k(f)$.

By separating the estimated signal component using the CLEAN technique, the remaining signal components can be further processed. The specific steps of this algorithm can be found in [26].

The GCPF algorithm can quickly analyze cubic phases of multicomponent signals. However, cross terms of the kernel function exist, which complicates the detection of the actual signal component from the transform domain spectrum. Although numerous studies have focused on suppressing these cross terms [26], false estimations still occur in certain cases. Moreover, under low signal-to-noise (SNR) ratio conditions, the estimation accuracy remains unsatisfactory.

b) SR with compact dictionary: The SR algorithms are commonly used for multicomponent signal analysis. They are not troubled by cross terms and have better noise robustness. SR is realized by solving the following optimization:

$$\hat{\vec{\sigma}} = \arg\min_{\vec{\sigma}} \|\mathbf{E} - \boldsymbol{\Psi}\vec{\sigma}\|_2 + \|\vec{\sigma}\|_1 \tag{23}$$

where $\mathbf{E} \in \mathcal{R}_{M\times 1}$ is the multicomponent signal, $\vec{\sigma} \in \mathcal{R}_{K_C\times 1}$ is the sparse coefficient vector, $\boldsymbol{\Psi} \in \mathcal{R}_{M\times K_C}$ is the dictionary, and its kth column is expressed as follows:

$$\vec{\mathbf{v}}_k(m+1) = \exp\left(j2\pi\left(\frac{\xi_{d,1,k}m}{\text{prf}} + \frac{\xi_{d,2,k}m^2}{\text{prf}^2} + \frac{\xi_{d,3,k}m^3}{\text{prf}^3}\right)\right) \tag{24}$$

where $\{\xi_{d,1,k}, \xi_{d,2,k}, \xi_{d,3,k}\}$ are the candidate coefficients, $k = 1, 2, \ldots, K_D$, $m = 0, 1, \ldots, M-1$, and $\vec{\mathbf{v}}_k \in \mathcal{R}_{M\times 1}$ is called the kth atom.

Since the accuracy of SR is heavily dependent on the completeness of the dictionary, it is crucial to select appropriate candidate coefficients. In traditional 2-D sparse imaging, the value range for the linear phase coefficients $\xi_{d,1,k}$ can be determined according to the imaging width [24], [38]. However, the value ranges for higher order phase coefficients cannot be determined due to the absence of prior information. Meanwhile, the dictionary should be streamlined to ensure efficiency.

Here, we use the coarse estimation in Section III-B to provide necessary prior information. Given that the GCPF algorithm usually achieves higher estimation accuracy for low-order phase coefficients [26], we adopt distinct approaches to determine value ranges for phase coefficients of different orders, referred to as the POHP strategy.

1) For linear phase coefficients, their coarse estimation accuracy is relatively high. We select candidate coefficients around the coarse estimation values, instead of uniformly sampling across the entire range. By discarding a large number of samples in regions where no coarse estimation values are present, the dictionary is streamlined. The specific selection process is given in Algorithm 1(a).
2) For quadratic and cubic phase coefficients, their coarse estimation accuracy is limited. We only use the coarse estimation values to determine the approximate value ranges of candidate coefficients. Then, we calculate the distribution density of the coarse estimation values, and the locations where the density is greater than the threshold are selected as candidate coefficients. This nonuniform sampling approach also makes the dictionary more compact. The specific selection process for quadratic phase coefficients is given in Algorithm 1(b), and the process for cubic phase coefficients is similar.

In Algorithm 1, K_C is the number of coarse estimates. $K_{D,1}$ and $K_{D,2}$ are respectively the numbers of the selected coefficients for linear and quadratic phases. The coefficients for cubic phases are selected with the interval $\Delta\xi_3 = 10^{-3}$, and its number is $K_{D,3}$. We combine coefficients for different order phases together, and the total number K_D equals $K_{D,1} \times K_{D,2} \times K_{D,3}$. The dictionary can be further established with these coefficients by (24).

Algorithm 1 Selection of Candidate Coefficients in Dictionary

a) Selection for linear phases

1: **Input:** The coarse estimates: $\{\hat{\xi}_{c,1,k}\}, k = 1, 2, \ldots, K_C$; the minimum sampling interval: $\Delta\xi_1 = 10^{-1}$.
2: **Output:** The selected candidate coefficients: $\{\xi_{d,1,k}\}, k = 1, 2, \ldots, K_{D,1}$.

3: Arrange $\{\hat{\xi}_{c,1,k}\}$ in ascending order to obtain $\{\xi_{a,1,k}\}$.
4: Initialize $k_1 = 1, \xi_{d,1,k_1} = \xi_{a,1,1} - \Delta\xi_1$.
5: For k from 2 to K_C do:
5.1: Calculate $\Delta\xi = \xi_{a,1,k} - \xi_{a,1,k-1}$.
5.2: If $\Delta\xi > 3\Delta\xi_1$, the candidate coefficients are selected in the vicinity of $\xi_{a,1,k-1}$ and $\xi_{a,1,k}$:

$$\xi_{d,1,k_1+i} = \left\{\xi_{a,1,k-1}, \xi_{a,1,k-1} + \Delta\xi_1, \xi_{a,1,k} - \Delta\xi_1\right\}, i = 1, 2, 3;$$

Else the candidate coefficients are uniformly sampled across the range between $\xi_{a,1,k-1}$ and $\xi_{a,1,k}$:

$$\xi_{d,1,k_1+i} = \left\{\xi_{a,1,k-1} + (i-1)\cdot\Delta\xi/3\right\}, i = 1, 2, 3.$$

5.3: $k_1 = k_1 + 3$.
6: $\xi_{d,1,k_1+1} = \xi_{a,1,K_C}, \xi_{d,1,k_1+2} = \xi_{a,1,K_C} + \Delta\xi_1$.

b) Selection for quadratic phases

1: **Input:** The coarse estimates: $\{\hat{\xi}_{c,2,k}\}, k = 1, 2, \ldots, K_C$; the minimum sampling interval: $\Delta\xi_2 = 10^{-2}$, the threshold $Thr_{den} = 0.4$.
2: **Output:** The selected candidate coefficients: $\{\xi_{d,2,k}\}, k = 1, 2, \ldots, K_{D,2}$.

3: Calculate $N_{sco} = \lceil(\max(\hat{\xi}_{c,2,k}) - \min(\hat{\xi}_{c,2,k}))/\Delta\xi_2\rceil + 8$, where $\lceil\cdot\rceil$ is the ceil function.
4: Determine search range $\xi = \xi_{\min} + (n_{sco} - 1)\cdot\Delta\xi_2$, where $n_{sco} = 1, 2, \ldots, N_{sco}$, and $\xi_{\min} = \min(\hat{\xi}_{c,2,k}) - 4\Delta\xi_2$
5: Calculate the distribution density:

$$f_{den}(\xi) = \sum_{k=1}^{K_C} \exp\left(-\frac{\left(\xi - \hat{\xi}_{c,2,k}\right)^2}{32(\Delta\xi_2)^2}\right).$$

6: Select ξ which satisfy $f_{den}(\xi) > Thr_{den}$ as $\{\xi_{d,2,k}\}$.

After the dictionary is determined, we solve the optimization (23) using SR algorithms, such as the approximate message passing (AMP) algorithm [39]. Based on the dictionary atoms corresponding to nonzero values of the vector $\vec{\sigma}_o$, the linear, quadratic, and cubic phase coefficients of strong scatterers in this selected range cell can be determined.

After the multicomponent signal decomposition for all the selected range cells is completed, we can obtain phase coefficients $\{\xi_{0,k}, \xi_{1,k}, \xi_{2,k}, \xi_{3,k}\}$ of strong scatterers in the total radar echo ($k = 1, 2, \ldots, K$).

C. Target Parameter Determination Based on Alternating Iterative Strategy

After the phase coefficients $\{\xi_{0,k}, \xi_{1,k}, \xi_{2,k}, \xi_{3,k}\}$ are estimated, we establish the overdetermined system of equations [see (17)] about target parameters $\Phi = \{\alpha, \beta, \omega, x_k, y_k, z_k\}$. This equation system can be expressed as an optimization problem by minimizing the residuals, and its objective function

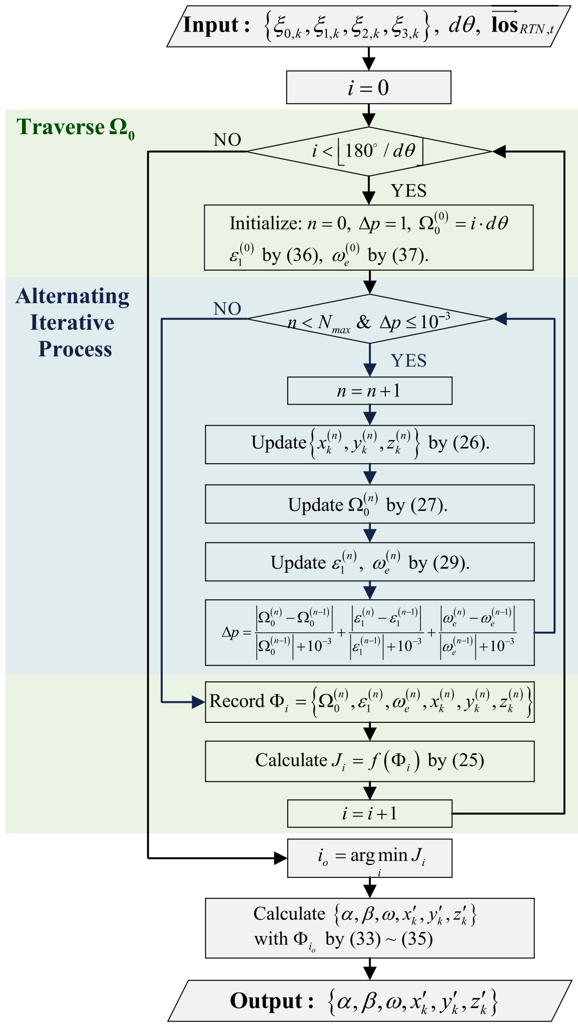

Fig. 5. Flowchart of the target parameter determination based on alternating iterative strategy.

is in the following equation:

$$f(\Phi)=\frac{1}{K}\sum_{k=1}^{K}\left\|\begin{bmatrix}\xi_{0,k}\\ \xi_{1,k}\\ \xi_{2,k}\\ \xi_{3,k}\end{bmatrix}+\frac{2f_c}{c}\mathbf{H}(\alpha,\beta,\omega)\begin{bmatrix}x_k\\ y_k\\ z_k\end{bmatrix}\right\|_2. \tag{25}$$

The optimization is high-dimensional and complex, and its optimal solution cannot be directly derived. It is also difficult for the conventional optimization algorithms, such as the particle swarm optimization (PSO) algorithm [40], to find the global optimum efficiently and accurately.

We introduce the alternating iterative strategy and derive the iterative formulas for location parameters $\{x_k, y_k, z_k\}$ and intermediate parameters $\{\Omega_0, \varepsilon_1, \omega_e\}$, which significantly reduces the search dimension. The flowchart of the target parameter determination is given in Fig. 5, where $q_{\max}$ is the maximum iteration number, $d\theta$ is the search interval of Ω_0, and $\lfloor\cdot\rfloor$ is the floor function. The related operations are explained below.

1) 3-D ISAR Imaging in the Nth Iteration: The iterative formula for 3-D location parameters of scatterers is represented as follows:

$$\begin{bmatrix}x_k^{(n)}\\ y_k^{(n)}\\ z_k^{(n)}\end{bmatrix}=-\frac{c}{2f_c}\cdot\left(\mathbf{H}\left(\Omega_0^{(n-1)},\varepsilon_1^{(n-1)},\omega_e^{(n-1)}\right)\Big\backslash\Xi_{\forall,k}\right) \tag{26}$$

where $\{x_k^{(n)}, y_k^{(n)}, z_k^{(n)}\}$ are location parameters in the XYZ system, $\Omega_0^{(n-1)}, \varepsilon_1^{(n-1)}, \omega_e^{(n-1)}$ are intermediate parameters estimated in the $(n-1)$th iteration, and $\Xi_{\forall,k}$ is the kth column of the phase coefficient matrix established by phase coefficients estimated in Section III-B. The left division for matrices is calculated by $\mathbf{A}\backslash\mathbf{B}=\mathbf{A}^{+}\mathbf{B}$, and $\mathbf{A}^{+}$ is the pseudoinverse of $\mathbf{A}$ obtained by the singular value decomposition (SVD).

2) Dynamic Estimation in the Nth Iteration: In the iterations, we estimate the intermediate parameters instead of the dynamic parameters $\{\alpha, \beta, \omega\}$. According to (13), we obtain the iterative formula for Ω_0

$$\Omega_0^{(n)}=\frac{1}{K}\sum_{k=1}^{K}\left(-\theta_k-\arcsin\left(\frac{c\xi_{0,k}}{2f_c\mu_k^{(n)}}\right)\right) \tag{27}$$

where θ_k is defined as (28), $\text{sgn}(\cdot)$ is the sign function, $\arcsin(\cdot)$ and $\arccos(\cdot)$ are the inverse trigonometric functions, and $\mu_k^{(n)}=((x_k^{(n)})^2+(z_k^{(n)})^2)^{1/2}$

$$\theta_k=\text{sgn}\left(\frac{z_k^{(n)}}{\mu_k^{(n)}}\right)\cdot\arccos\left(\frac{x_k^{(n)}}{\mu_k^{(n)}}\right). \tag{28}$$

According to (14), we obtain iterative formulas for ε_1 and ω_e

$$\begin{bmatrix}\varepsilon_1^{(n)}\\ \omega_e^{(n)}\end{bmatrix}=-\frac{c}{2f_c}\cdot\mathbf{\Gamma}_1\backslash\Xi_{2,\forall}$$

$$\mathbf{\Gamma}_1=\begin{bmatrix}x_1^{(n)}\cos -z_1^{(n)}\sin\Omega_0^{(n)} & -y_1^{(n)}\sin\Omega_0^{(n)}\\ x_2^{(n)}\cos\Omega_0^{(n)}-z_2^{(n)}\sin\Omega_0^{(n)} & -y_2^{(n)}\sin\Omega_0^{(n)}\\ \cdots & \cdots\\ x_K^{(n)}\cos\Omega_0^{(n)}-z_K^{(n)}\sin\Omega_0^{(n)} & -y_K^{(n)}\sin\Omega_0^{(n)}\end{bmatrix} \tag{29}$$

where $\Xi_{2,\forall}$ is the 2nd row of the phase coefficient matrix. Considering that the estimation accuracy of lower-order phase coefficients is usually higher, we do not use (15) and (16) in dynamic estimation.

In a special case ($|\Omega_0^{(n)}|<1^\circ$, i.e., $\sin\Omega_0^{(n)}\approx 0$), ω_e needs to be further determined by quadratic phase coefficients to ensure its accuracy. Its iterative formula is derived as follows:

$$\omega_e^{(n)}=-\frac{c}{2f_c}\cdot\mathbf{\Gamma}_{2,1}\backslash\left(\Xi_{3,\forall}+\mathbf{\Gamma}_{2,2}\right)$$

$$\mathbf{\Gamma}_{2,1}=\begin{bmatrix}-\varepsilon_1^{(n)}y_1^{(n)}\cos\Omega_0^{(n)}\\ -\varepsilon_1^{(n)}y_2^{(n)}\cos\Omega_0^{(n)}\\ \cdots\\ -\varepsilon_1^{(n)}y_K^{(n)}\cos\Omega_0^{(n)}\end{bmatrix}$$

$$\mathbf{\Gamma}_{2,2}=\frac{1}{2}\begin{bmatrix}\left(\varepsilon_1^{(n)}\right)^2\cos\Omega_0^{(n)}z_1^{(n)}\\ \left(\varepsilon_1^{(n)}\right)^2\cos\Omega_0^{(n)}z_2^{(n)}\\ \cdots\\ \left(\varepsilon_1^{(n)}\right)^2\cos\Omega_0^{(n)}z_K^{(n)}\end{bmatrix}. \tag{30}$$

3) Target Parameter Determination in the RTN System: After the alternating iterative process, we further estimate the dynamic parameters and location parameters in the RTN system.

To estimate dynamic parameters from intermediate parameters $\{\Omega_0, \varepsilon_1, \omega_e\}$, we introduce two auxiliary vectors $\vec{\mathbf{v}}_1$ and $\vec{\mathbf{v}}_2$. $\vec{\mathbf{v}}_1$ is a normalized vector that satisfies $\vec{\mathbf{v}}_1^{\mathrm{T}}\overrightarrow{\mathbf{los}}_{\mathrm{RTN},0} = 0$, and $\vec{\mathbf{v}}_2 = \vec{\mathbf{v}}_1 \otimes \overrightarrow{\mathbf{los}}_{\mathrm{RTN},0}$. The direction of spin axis can be rewritten as follows:

$$\vec{\boldsymbol{\omega}} = \begin{bmatrix} \vec{\mathbf{v}}_1 & \vec{\mathbf{v}}_2 & \overrightarrow{\mathbf{los}}_{\mathrm{RTN},0} \end{bmatrix} \begin{bmatrix} \cos\phi \sin\Omega_0 \\ \sin\phi \sin\Omega_0 \\ \cos\Omega_0 \end{bmatrix}. \tag{31}$$

According to $\cos(\Omega_0 + \varepsilon_1 \cdot t) = \overrightarrow{\mathbf{los}}_{\mathrm{RTN},t}^{\mathrm{T}}\vec{\boldsymbol{\omega}}$, the angle ϕ can be estimated by the following equation:

$$\begin{aligned} \phi &= \frac{1}{M-1}\sum_{m=1}^{M-1}\left(\arcsin\left(\frac{\Delta\cos_t}{\eta_t \sin\Omega_0}\right) - \varphi_t\right) \\ \Delta\cos_t &= \cos\left(\Omega_0 + \varepsilon_1 t\right) - \cos\Omega_0 \overrightarrow{\mathbf{los}}_{\mathrm{RTN},t}^{\mathrm{T}}\overrightarrow{\mathbf{los}}_{\mathrm{RTN},0} \\ \varphi_t &= \operatorname{sgn}\left(\eta_{t,1}/\eta_t\right) \cdot \arccos\left(\eta_{t,2}/\eta_t\right) \end{aligned} \tag{32}$$

where $t = m/\mathrm{prf}, m = 0, 1, \ldots, M-1, \eta_{t,1} = \overrightarrow{\mathbf{los}}_{\mathrm{RTN},t}^{\mathrm{T}}\vec{\mathbf{v}}_1$, $\eta_{t,2} = \overrightarrow{\mathbf{los}}_{\mathrm{RTN},t}^{\mathrm{T}}\vec{\mathbf{v}}_2$, and $\eta_t = (\eta_{t,1}^2 + \eta_{t,2}^2)^{1/2}$. Then, the dynamic parameters α and β is estimated by the following equation:

$$\begin{aligned} \beta &= \arcsin\left(\vec{\boldsymbol{\omega}}(3)\right) \\ \alpha &= \operatorname{sgn}\left(\frac{\vec{\boldsymbol{\omega}}(2)}{\cos\beta}\right) \cdot \arccos\left(\frac{\vec{\boldsymbol{\omega}}(1)}{\cos\beta}\right). \end{aligned} \tag{33}$$

The spin velocity ω is $\omega_e + \varepsilon_2$, where ε_2 is estimated by the following equation:

$$\varepsilon_2 = \arccos\left(\frac{\overrightarrow{\mathbf{los}}_{\mathrm{RTN},t}^{\mathrm{T}}\overrightarrow{\mathbf{los}}_{\mathrm{RTN},0} - \cos\Omega_0\cos(\Omega_0 + \varepsilon_1 t)}{\sin\Omega_0 \sin\left(\Omega_0 + \varepsilon_1 t\right)}\right)\Bigg/ t. \tag{34}$$

Notably, location parameters in the XYZ system cannot reflect the target absolute attitude in orbit. We further obtain location parameters in the RTN system by coordinate transformation

$$\begin{bmatrix} x_k' \\ y_k' \\ z_k' \end{bmatrix} = \begin{bmatrix} \vec{\mathbf{x}} & \vec{\mathbf{y}} & \vec{\mathbf{z}} \end{bmatrix} \begin{bmatrix} x_k \\ y_k \\ z_k \end{bmatrix} \tag{35}$$

where the axis directions are obtained by (3).

4) Analysis of Search Range: In the 3-D imaging, the intermediate parameters are fixed. According to (26), the related equations for location parameters are linear, allowing location parameters to approach their optimal solutions without concern for local extrema.

In the dynamic estimation, the quadratic and cubic phase coefficients are not used; ε_1 and ω_e can approach their optimal solutions according to (29). However, the relationship between phase coefficients and Ω_0 is nonlinear, necessitating a global search to avoid local extremes. Here we give the variation curves of $f(\Phi)$ with respect to Ω_0, ε_1 and ω_e. In Fig. 6(a), Ω_0 traverses its possible search range, ε_1 and ω_e are fixed as their optimal solutions, and location parameters are calculated by (26). The local extremum of its curve is marked in red. Similarly, ε_1 and ω_e respectively traverse their possible search ranges in Fig. 6(b) and (c). We can find that the curves with respect to ε_1 and ω_e have no local extremum. We set the search range of Ω_0 as $[-90°, 90°]$ and the search interval as $10°$. Then, ε_1 and ω_e are initialized as follows:

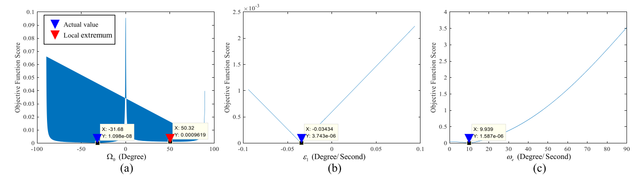

Fig. 6. Variation curves of the objective function with respect to intermediate parameters (Actual values: $\Omega_0 = -31.68°$, $\varepsilon_1 = -0.035°/\mathrm{s}$, $\omega_e = 9.939°/\mathrm{s}$). (a) Curve of Ω_0 from $-90°$ to $90°$. (b) Curve of ε_1 from $-\varepsilon_1^{(0)}$ to $\varepsilon_1^{(0)}$. (c) Curve of ω_e from $0°/\mathrm{s}$ to $90°/\mathrm{s}$.

TABLE I

RADAR AND IMAGING PARAMETERS

Radar Location	Hangzhou (120.2E,30.3N,0m)	
Radar Parameters	Center Frequency	10 GHz
	Bandwidth	1 GHz
	PRF	80 Hz
Imaging Parameters	2-D Imaging CPI	256 Pulses
	2-D Imaging Interval	128 Pulses
	Number of 2-D Images	5
	3-D Imaging CPI	512 Pulses

$$\varepsilon_1^{(0)} = \arccos\left(\overrightarrow{\mathbf{los}}_{\mathrm{RTN},t}^{\mathrm{T}}\overrightarrow{\mathbf{los}}_{\mathrm{RTN},0}\right)/t \tag{36}$$

$$\omega_e^{(0)} = \frac{1}{K}\sum_{k=1}^{K}\frac{\xi_{1,k}}{\xi_{0,k}}. \tag{37}$$

In summary, our algorithm requires only a 1-D search for Ω_0, whereas conventional optimization algorithms operate within a search space of $3K+3$ dimensions. By employing the alternating iterative strategy and using the intermediate parameters, we significantly decrease computational complexity.

IV. EXPERIMENTAL ANALYSIS

A. Experiment Based on Simple Point Model

In our experiments, the radar parameters are given in Table I. To satisfy the assumption of fixed dynamic parameters, the whole observation period is set within 10 s [7]. The simple point model (46 points) and its initial attitude in the RTN system are shown in Fig. 7(a). The radar observation angles are calculated in STK-11 software based on the actual orbital elements of Landsat-7. Their variation curves are given in Fig. 7(b) and (c).

The comparative methods are selected as follows.

1) For dynamic estimation, the imaging projection-based methods can be applied to uncooperative targets and share similar estimation performance [3], [7], [8], [9]. Among these methods, we choose the modified image sequence energy accumulation (MISEA) method [3] and the line-feature-based (LFB) method [7] as representative comparative approaches.

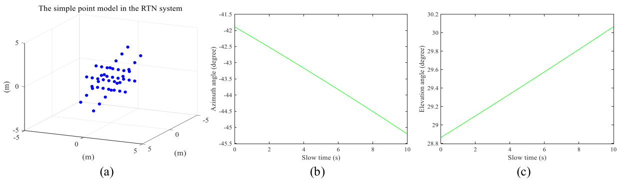

Fig. 7. Simple point model and radar observation angles. (a) Simple point model. (b) Variation curve of azimuth angles. (c) Variation curve of elevation angles.

2) For 3-D geometry reconstruction, among the imaging projection-based methods [3], [11], [12], [13], [14], [15], only the MISEA method [3] is applicable to spinning spacecraft. Besides, the orthogonal-factorization-based (OFB) method [10] is also chosen as our comparative approach.

For our 3-D ISAR imaging, the imaging CPI is set to 512 pulses. For comparative methods based on 2-D images, an ISAR image sequence, whose imaging parameters are listed in Table I, is generated. These images are obtained from the simulated echoes using the Range-Doppler (R-D) and autofocusing algorithms [23]. Their cross ranges are further scaled by the dynamic estimation method [7]

$$\Delta d = \frac{c}{2 f_c \hat{\omega}_e T_{\mathrm{CPI}}} \tag{38}$$

where $\hat{\omega}_e$ is the effective spin velocity estimated by dynamic estimation, and T_{CPI} is the 2-D imaging CPI.

1) Experiments for Different Spin Velocities: In this experiment, the spin velocity varies from 0.0524 to 0.3491 rad/s. The direction of the spin axis is set as $[0.341, 0.029, -0.939]^{\mathrm{T}}$ (The initial angle Ω_0 between the spin axis and the radar LOS vector is 75°). The SNR is set to 10 dB.

First, we analyze the impact of target spin on 2-D ISAR imaging. Some images under different spin velocities are given in Fig. 8, and their degrees of defocusing are assessed by image entropy (IE) in the following equation:

$$\mathrm{IE} = -\sum_{m=1}^{M}\sum_{n=1}^{N} \frac{\left|I_{m,n}\right|^2}{\mathrm{Ene}} \cdot \ln\left(\frac{\left|I_{m,n}\right|^2}{\mathrm{Ene}}\right)$$
$$\mathrm{Ene} = \sum_{m=1}^{M}\sum_{n=1}^{N}\left|I_{m,n}\right|^2 \tag{39}$$

where $I_{m,n}$ represents the pixel at the mth row and nth column of the image I. The defocusing image usually has a larger entropy. When the spin velocity exceeds 0.1745 rad/s, the ISAR images exhibit significant defocusing, making it difficult to accurately extract the geometric features used by some existing methods. For illustration, in Fig. 8(a), the line features used in the LFB method and key point features used in the OFB method are respectively marked in green and blue. In Fig. 8(d) and (e), some target scatterers, which are circled in purple, are defocusing. In the OFB method, key points can hardly be extracted from Fig. 8(d) and (e), and the LFB method fails to extract line features from Fig. 8(e). The failure of extracting these geometric features renders these methods ineffective.

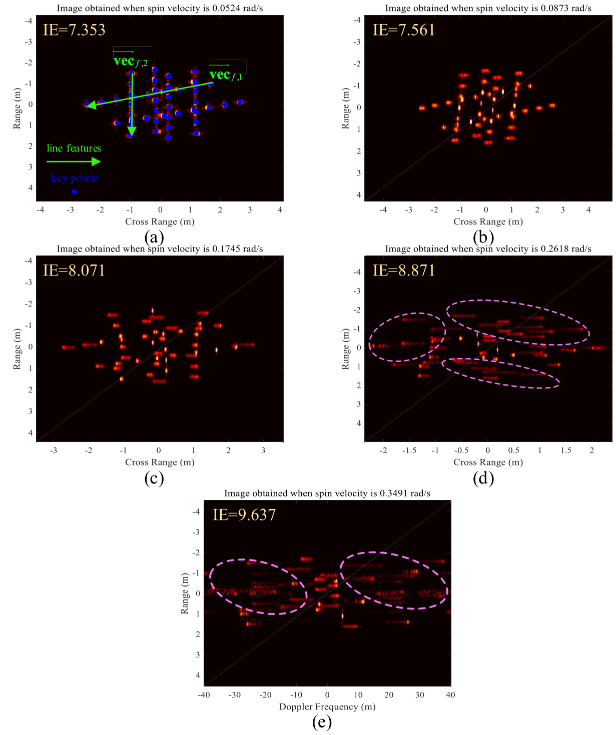

Fig. 8. 2-D ISAR images under different spin velocities (SNR: 10 dB). (a) 0.0524, (b) 0.0873, (c) 0.1745, (d) 0.2618, and (e) 0.3491 rad/s. Their degrees of defocusing are assessed by IE. In (a), line features utilized in existing methods are marked in green, and key points are marked in blue. In (d) and (e) the scatterers circled in purple are defocusing, and their energy is dispersed in the cross range.

Then, we compare the accuracies of dynamic estimation and 3-D reconstruction across different methods.

The 3-D reconstruction results at different spin velocities are shown in Fig. 9. To quantitatively evaluate their accuracy, an index is defined as follows:

$$Er_R = \frac{1}{K_r}\sum_{k=1}^{K_r}\min_i\left\|\vec{\mathbf{p}}_{a,i} - \vec{\mathbf{p}}_{r,k}\right\|_2 + \frac{1}{K_a}\sum_{k=1}^{K_a}\min_i\left\|\vec{\mathbf{p}}_{r,i} - \vec{\mathbf{p}}_{a,k}\right\|_2 \tag{40}$$

where $\{\vec{\mathbf{p}}_{a,i}\}$ is 3-D locations of the actual target points, $K_a = 46$, $\{\vec{\mathbf{p}}_{r,i}\}$ is 3-D locations of the reconstructed target points, K_r is the number of reconstructed points.

The accuracy of dynamic estimation is assessed by errors of spin velocity and spin axis direction. These indices obtained under different spin velocities are presented in Table II.

For dynamic estimation, the errors obtained by our method are similar to those of the existing methods when the spin velocity is 0.0524 rad/s. However, as the spin velocity increases, the errors obtained by our method are obviously lower. For 3-D reconstruction, our method consistently achieves lower errors compared to the existing methods across all spin velocities. These improvements may be attributed to the following factors. On the one hand, our method does not rely on the quality of 2-D ISAR images, and their defocusing has no impact on our method. On the other hand, the high-order phases of radar echoes, which degrade the quality

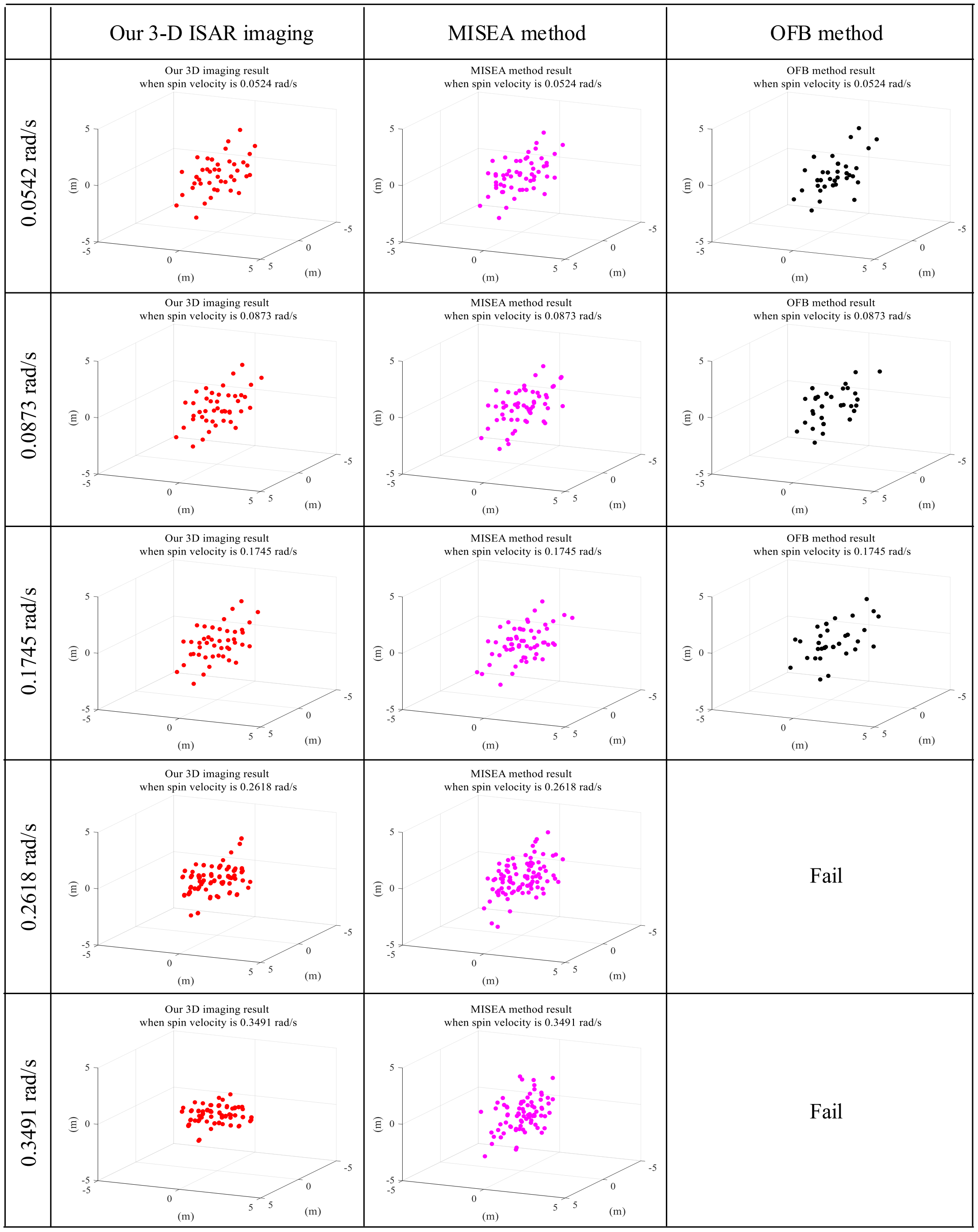

Fig. 9. 3-D reconstruction results under different spin velocities. Images in the 1st row are results obtained by our 3-D ISAR imaging. Images in the 2nd row are results obtained by the MISEA method. Images in the 3rd row are results obtained by the OFB method. The OFB method is ineffective when the spin velocity is 0.2618 or 0.3491 rad/s.

of 2-D imaging, are leveraged by our method to provide additional information. Although the MISEA method is also effective when the spin velocity is high, its estimation and reconstruction accuracies drop significantly.

Finally, we provide a detailed analysis of the impact of spin velocity on our method. Within the same CPI, the difference between two adjacent echoes is greater at a higher spin velocity, providing richer information for estimation. Therefore, our method achieves lower errors at a spin velocity of 0.0873 rad/s compared to 0.0524 rad/s. When the spin velocity is higher than 0.1745 rad/s, the rotational angle during the CPI exceeds $\pi/3$. The approximation of trigonometric functions in (11) introduces errors, and the accuracies of dynamic estimation and 3-D reconstruction drop. What is more, Doppler ambiguity is a common issue in radar imaging. The condition for imaging without Doppler ambiguity is $\max_k |(2f_c/c) \cdot \vec{\mathbf{h}}_1 \vec{\mathbf{p}}_k| < \text{PRF}/2$. In this experiment, Doppler ambiguity arises when the spin velocity is above 0.2855 rad/s. As shown in Table II, when the spin velocity is 0.3491 rad/s, Doppler ambiguity leads to a serious decline in accuracy. To avoid the range and

TABLE II

ERRORS OF DYNAMIC ESTIMATION AND 3-D RECONSTRUCTION UNDER DIFFERENT SPIN VELOCITIES. (a) ERROR OF SPIN VELOCITY ESTIMATION (rad/s). (b) ERROR OF SPIN AXIS DIRECTION ESTIMATION (DEGREE). (c) ERROR OF 3-D RECONSTRUCTION (METER)

(a)

Spin velocity (rad/s)	0.0524	0.0873	0.1745	0.2618	0.3491
Our Method	2×10^{-4}	2×10^{-4}	2×10^{-4}	4×10^{-4}	16×10^{-4}
MISEA Method	2×10^{-4}	4×10^{-4}	9×10^{-4}	35×10^{-4}	51×10^{-4}
LFB Method	2×10^{-4}	4×10^{-4}	17×10^{-4}	48×10^{-4}	fail

(b)

Spin velocity (rad/s)	0.0524	0.0873	0.1745	0.2618	0.3491
Our Method	1.649	1.086	1.164	2.907	8.546
MISEA Method	1.978	2.738	3.964	13.569	21.897
LFB Method	2.112	4.251	5.848	17.995	fail

(c)

Spin velocity (rad/s)	0.0524	0.0873	0.1745	0.2618	0.3491
Our Method	0.494	0.436	0.446	0.586	0.965
MISEA Method	0.512	0.549	0.625	0.873	1.265
OFB Method	0.784	0.923	1.453	fail	fail

TABLE III

ERRORS OF DYNAMIC ESTIMATION AND 3-D RECONSTRUCTION UNDER DIFFERENT SPIN AXES. (a) DIFFERENT SPIN AXES IN SECTION IV-A2. (b) ERROR OF SPIN VELOCITY ESTIMATION (rad/s). (c) ERROR OF SPIN AXIS DIRECTION ESTIMATION (DEGREE). (d) ERROR OF 3-D RECONSTRUCTION (METER)

(a)

Ω_0 (degrees)	0	45	75	90
Direction of spin axis	[0.632; -0.599; 0.491]	[0.819; -0.518; -0.247]	[0.341; 0.029; -0.939]	[0.526; -0.133; -0.840]

(b)

Ω_0 (degrees)	0	45	75	90
Our method	3×10^{-4}	2×10^{-4}	2×10^{-4}	2×10^{-4}
MISEA method	4×10^{-4}	4×10^{-4}	4×10^{-4}	4×10^{-4}
LFB method	4×10^{-4}	4×10^{-4}	4×10^{-4}	5×10^{-4}

(c)

Ω_0 (degrees)	0	45	75	90
Our method	1.835	1.152	1.086	0.988
MISEA method	3.532	2.511	2.738	2.917
LFB method	4.355	4.016	4.251	4.986

(d)

Ω_0 (degrees)	0	45	75	90
Our method	0.502	0.445	0.436	0.402
MISEA method	0.598	0.566	0.549	0.613
Existing method	0.910	0.875	0.923	0.985

Doppler ambiguity at the same time, careful selection of PRF is crucial [41].

2) Experiments for Different Spin Axes: In this experiment, the initial angle Ω_0 between the spin axis and the radar LOS vector varies from 0° to 90°. Then, the direction of spin axis is listed in Table III(a). The spin velocity is set to 0.0873 rad/s.

TABLE IV

ERRORS OF DYNAMIC ESTIMATION AND 3-D RECONSTRUCTION UNDER DIFFERENT SNRS. (a) ERROR OF SPIN VELOCITY ESTIMATION (rad/s). (b) ERROR OF SPIN AXIS DIRECTION ESTIMATION (DEGREE). (c) ERROR OF 3-D RECONSTRUCTION (METER)

(a)

SNR (dB)	-5	0	5	10
Our method	4×10^{-4}	2×10^{-4}	2×10^{-4}	2×10^{-4}
MISEA method	5×10^{-4}	4×10^{-4}	4×10^{-4}	4×10^{-4}
LFB method	6×10^{-4}	4×10^{-4}	4×10^{-4}	4×10^{-4}

(b)

SNR (dB)	-5	0	5	10
Our method	3.672	2.025	1.334	1.086
MISEA method	4.025	3.395	3.013	2.738
LFB method	6.533	4.910	4.585	4.551

(c)

SNR (dB)	-5	0	5	10
Our method	0.912	0.611	0.492	0.436
MISEA method	0.993	0.717	0.604	0.549
LFB method	1.928	1.758	1.397	1.103

The SNR is set to 10 dB. The errors of dynamic estimation and 3-D reconstruction are presented in Table III(b)~(d). Under all spin axes set in this experiment, our method achieves higher accuracy than the comparative methods. The experimental results indicate that our 3-D ISAR imaging model is robust to various spin axes.

3) Experiments for Different SNRs: Considering that the SNRs of radar echoes in realistic space observation scenarios are around 0 dB, the SNR in this experiment varies from -5 to 10 dB. The spin velocity is set to 0.0873 rad/s, and the direction of spin axis is set as $[0.341, 0.029, -0.939]^{\mathrm{T}}$. The errors of dynamic estimation and 3-D reconstruction are presented in Table IV. Because of the noise sensitivity of high order phases, the errors of our method significantly increase when the SNR is -5 dB. However, the accuracy of our method remains superior to the existing methods across all tested SNRs.

4) Analysis of Phase Coefficient Estimation: The reliability and superiority of our phase coefficient estimation are analyzed. In this section, the spin velocity is set to 0.0524 rad/s. The direction of the spin axis is set as $[0.341, 0.029, -0.939]^{\mathrm{T}}$.

a) Compare the noise robustness of GCPF and SR: The estimation accuracy of the coarse estimation based on GCPF and the precise estimation based on SR are analyzed. The phase coefficient estimation error is defined as follows:

$$Er_{\xi} = \frac{\sum_{k=1}^{K_e} \min_i \left\| \vec{\xi}_i - \hat{\vec{\xi}}_k \right\|_2}{K_e} + \frac{\sum_{k=1}^{K} \min_i \left\| \hat{\vec{\xi}}_i - \vec{\xi}_k \right\|_2}{K} \tag{41}$$

where $\vec{\xi}_i = [\xi_{1,i}, \xi_{2,i}, \xi_{3,i}]^{\mathrm{T}}$ are the actual phase coefficients $(i = 1, 2, \ldots, K)$; $\hat{\vec{\xi}}_i = [\hat{\xi}_{1,i}, \hat{\xi}_{2,i}, \hat{\xi}_{3,i}]^{\mathrm{T}}$ are the estimated phase coefficients $(i = 1, 2, \ldots, K_e)$. The errors of coarse and precise estimation under different SNRs are shown in Table V. When the SNR is -5 dB, a large number of false scatterers are extracted by the coarse estimation, resulting in a sharp decline in accuracy. The precise estimation effectively filters these

TABLE V
ERRORS OF PHASE COEFFICIENT ESTIMATION UNDER DIFFERENT SNRS

SNR (dB)	-5	0	5	10
Coarse estimation based on GCPF	0.425	0.145	0.111	0.098
Precise estimation based on SR	0.197	0.102	0.098	0.093

TABLE VI
SELECTION RANGE AND INTERVAL OF CONVENTICAL DICTIONARY

	Selection range	Selection interval
Linear phase	[-80,80]	10^{-1}
Quadratic phase	[-1.51,1.51]	10^{-2}
Cubic phase	[-0.152,0.152]	10^{-3}

false scatterers by leveraging the sparsity of actual scatterers. As shown in Table V, the precise estimation improves the noise-robustness of phase coefficient estimation.

b) Analyze the size and completeness of dictionary: First, we compare the size of our compact dictionary with that of the conventical overcomplete dictionary.

The conventical dictionary selects its candidate phase coefficients at fixed intervals within specific ranges. The selection ranges for its quadratic and cubic phase coefficients can hardly be determined in practice. In this experiment, we determine the ranges according to their actual values. The selection intervals are consistent with the minimum sampling intervals of our compact dictionary. The detailed selection process is described as follows. The specific ranges and intervals are listed in Table VI. Its size is around $M \times K_{\text{over}}$ ($M = 512$, $K_{\text{over}} = 7.34 \times 10^7$). When stored as a "double" data type, the storage requirement of conventical dictionary exceeds 200 GB. Its size exceeds the storage capacity of our computer storage, resulting in the failure of SR. Therefore, the coarse estimation is indispensable for designing a suitable dictionary for SR.

Our compact dictionaries vary in size for different range cells, with an average size of approximately $M \times K_{\text{com}}$ ($K_{\text{com}} = 3.5 \times 10^3$). Its storage requirement is around 12 MB. We significantly decrease the dictionary size by using the POHP strategy. The computational complexity of the AMP algorithm is $O(MK)$, and SR using our compact dictionary has markedly higher computational efficiency.

Then, we evaluate the completeness of dictionaries. In our application, the dictionary is considered complete if its candidate phase coefficients encompass all the actual phase coefficients. The completeness of the conventional dictionary is measured by the following equation:

$$\text{comp}_1 = \left(\frac{1}{K} \sum_{k=1}^{K} \min_i \left\| \vec{\boldsymbol{\xi}}_{d,i} - \vec{\boldsymbol{\xi}}_k \right\|_2 \right)^{-1} \tag{42}$$

where $\vec{\boldsymbol{\xi}}_{d,i} = [\xi_{d,1,i}, \xi_{d,2,i}, \xi_{d,3,i}]^{\mathrm{T}}$ are the candidate phase coefficients in the conventional dictionary, and $\vec{\boldsymbol{\xi}}_k$ are the actual phase coefficients. In this experiment, comp_1 equals 20.45. The completeness of our compact dictionaries is

TABLE VII
STORAGE REQUIREMENT AND COMPLETENESS OF DICTIONARIES

	Storage requirement	Completeness
Our compact dictionary	12 MB	19.12~20.45
Conventional dictionary	200 GB	20.45

TABLE VIII
COMPUTATIONAL TIME OF TARGET PARAMETER DETERMINATION

Our alternating iterative algorithm	3.2 s
PSO algorithm	103.5 s

measured by the following equation:

$$\text{comp}_2 = \left(\frac{1}{K} \sum_{n=1}^{N} \sum_{k=1}^{K_n} \min_i \left\| \vec{\boldsymbol{\xi}}_{n,d,i} - \vec{\boldsymbol{\xi}}_{n,k} \right\|_2 \right)^{-1} \tag{43}$$

where $\vec{\boldsymbol{\xi}}_{n,k} = [\xi_{n,1,k}, \xi_{n,2,k}, \xi_{n,3,k}]^{\mathrm{T}}$ are the actual phase coefficients of scatterers in the nth range cell ($i = 1, 2, \ldots, K_n$), and $\vec{\boldsymbol{\xi}}_{n,d,i} = [\xi_{n,d,1,i}, \xi_{n,d,2,i}, \xi_{n,d,3,i}]^{\mathrm{T}}$ are the candidate phase coefficients in the compact dictionary for the nth range cell. Although the coarse estimation based on GCPF is sensitive to noise, the false scatterers introduced under low SNRs have little impact on the completeness of the designed dictionary. When the SNR is between -5 and 10 dB, the completeness of our compact dictionary is between 19.12 and 20.45, which is comparable to the completeness of conventional dictionary. As summarized in Table VII, our dictionary effectively achieves a balance between lightweight design and completeness.

5) Analysis of Target Parameter Determination: As analyzed in Section III-C4, our alternating iterative algorithm converts the high dimensional optimization into a 1-D search problem. This section presents the computational time required for our alternating iterative algorithm and the conventional optimization algorithm (PSO algorithm) to identify the global optimum. In this experiment, our algorithm traverses Ω_0 within $[-90°, 90°]$ at an interval of 10°. For each Ω_0, our algorithm converges within 3~6 iterations. The PSO algorithm, configured with a particle number of 30, converges after around 300 iterations. As shown in Table VIII, the computational time of the PSO algorithm is approximately 30 times greater than that of our algorithm.

B. Experiment Based on Electromagnetic Simulation

To generate more realistic radar data, the electromagnetic simulation is conducted in EMSS FEKO 7.0 software using the computer-aided design (CAD) model of Landsat-7. The radar parameters and observation angles are consistent with those in Section IV-A, which replicates a realistic space observation scenario. The direction of spin axis is $[0.341, 0.029, -0.939]^{\mathrm{T}}$, and the spin velocity is 0.0873 rad/s. The SNR is set to 0 dB. The errors of dynamic estimation and 3-D reconstruction obtained by our method and the existing methods are given in Table IX. Our method outperforms the existing methods in terms of accuracies.

TABLE IX

ERRORS OF DYNAMIC ESTIMATION AND 3-D RECONSTRUCTION IN EXPERIMENT BASED ON ELECTROMAGNETIC SIMULATION

Error of spin velocity (rad/s)	Our dynamic estimation	2×10^{-4}
	MISEA method	4×10^{-4}
	LFB method	5×10^{-4}
Error of spin axis direction (degree)	Our dynamic estimation	2.347
	MISEA method	3.811
	LFB method	5.123
Error of 3D reconstruction (meter)	Our 3D ISAR imaging	0.712
	MISEA method	0.803
	OFB method	1.221

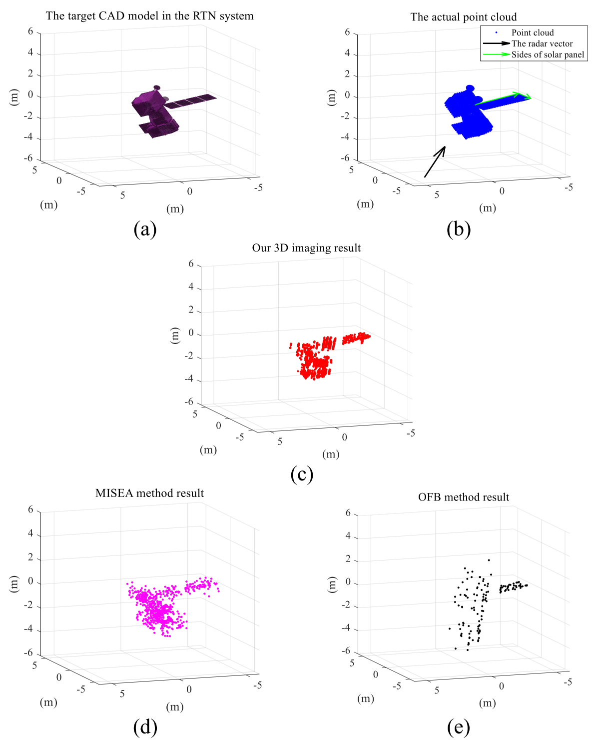

Fig. 10. Actual 3-D model and reconstructed point cloud of Landsat-7. (a) CAD model. (b) Actual point cloud. (c) Point cloud reconstructed by our 3-D ISAR imaging. (d) Point cloud reconstructed by the MISEA method. (e) Point cloud reconstructed by the OFB method. In (b), the black vector is the initial LOS, and the two green vectors are the sides of the solar panel.

During the electromagnetic simulation, some scatterers vary or even slip as the radar LOS changes (angular glint phenomenon) [42]. This leads to a slight decrease in accuracy of dynamic estimation compared with IV-A. Besides, due to occlusion among scatterers, the accuracy of 3-D reconstruction drops significantly. As shown in Fig. 10(b)–(d), some target scatterers are not irradiated by radar, therefore, cannot be reconstructed. The occlusion issue arises in most reconstruction methods due to lack of diverse observation views.

Notably, as long as there exist more than four strong scatterers, the equation system (17) is overdetermined ($4K > 3K + 3$), and our framework is effective. Consequently, although some scatterers may be missing or incorrectly extracted, the dynamic parameters can still achieve high estimation accuracy.

Moreover, the incomplete reconstruction results can also reflect some target structural information. For example, two adjacent sides of the solar panel can be extracted from the point clouds, which allows further estimating their sizes and initial attitudes [15]. The estimation errors are given in Table X. Our 3-D ISAR imaging results can provide more accurate features compared to the point cloud reconstructed by the factorization-based method.

TABLE X

ERRORS OF SIZE AND ATTITUDE ESTIMATED FROM 3-D RECONSTRUCTION RESULTS

3-D reconstruction method		Our method	MISEA method	OFB method
Size estimation error	Side 1	0.24 m	0.26 m	0.74 m
	Side 2	0.11 m	0.28 m	0.14 m
Attitude estimation error	Side 1	1.72°	2.04°	4.97°
	Side 2	2.52°	3.36°	6.72°

C. Some Discussions

The experimental results above demonstrate the performance advantages of our framework. However, several challenges remain to be addressed to enhance its practical application. As discussed in IV-B, the angular glint phenomenon and occlusion among scatterers negatively affect the accuracies of dynamic estimation and 3-D reconstruction. These issues are particularly prevalent in both existing methods and our framework.

To alleviate the influence of the angular glint phenomenon, we can model the scatterer amplitude as time-varying functions during phase coefficient estimation. For example, Wang et al. [43] model the time-varying amplitude as a slowly varying cosine function, which effectively improves estimation accuracy in cases of scatterer variation and slipping. Correspondingly, our precise estimation based on SR should incorporate parameters of cosine functions into the dictionary, which will inevitably increase the dictionary size. Further research is necessary to optimize the computational efficiency of SR while maintaining its accuracy.

To tackle the occlusion issue, we can use multiple spaceborne radars for simultaneous observation to obtain abundant views [44]. In [45], the LOS vectors of spaceborne radars are transformed into the target RTN coordinate system using GPS information and the tracking trajectory of payload satellites. With these LOS vectors, our 3-D ISAR imaging model can also be applied to spaceborne radars. After 3-D ISAR imaging is performed for each radar, point cloud fusion algorithms should be developed to achieve complete 3-D reconstruction.

V. CONCLUSION

In this article, we propose a 3-D ISAR imaging and dynamic estimation framework based on signal phase analysis for noncooperative spinning spacecraft. Our two-step framework estimates the 3-D locations of scatterers, spin velocity, and direction of spin axis directly from cubic phase coefficients of radar echoes, thereby avoiding issues caused by image defocusing and cross-range scaling errors. In phase coefficient estimation, the coarse estimation based on the GCPF algorithm enables the design of more compact dictionaries, enhancing the computational efficiency of SR. Meanwhile, the precise estimation based on SR significantly improves the estimation accuracy under low SNRs. In target parameter determination, we introduce the alternating iterative strategy to quickly find the global optimum value of the complex equation system. The experimental results validate the superiority of our framework compared to existing methods. Our performance advantages are increasingly evident as spin velocity increases. However, challenges such as variations in scatterer amplitudes and occlusion among scatterers remain to be addressed to promote the practical application.

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